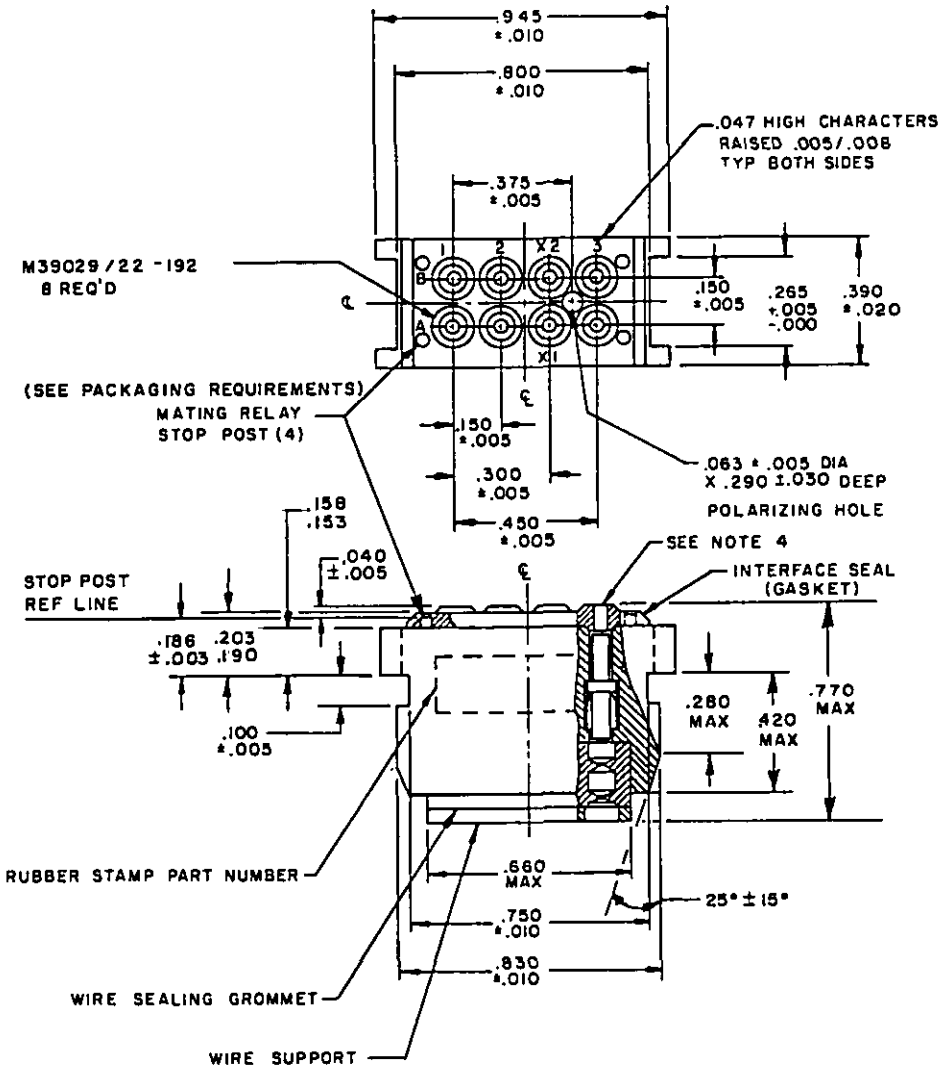
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, SOCKET MOUNT

This specification sheet is approved for use by the Department of the Air Force, and is available for use within all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sockets described herein shall consist of this specification and the latest issue of MIL-R-6106.

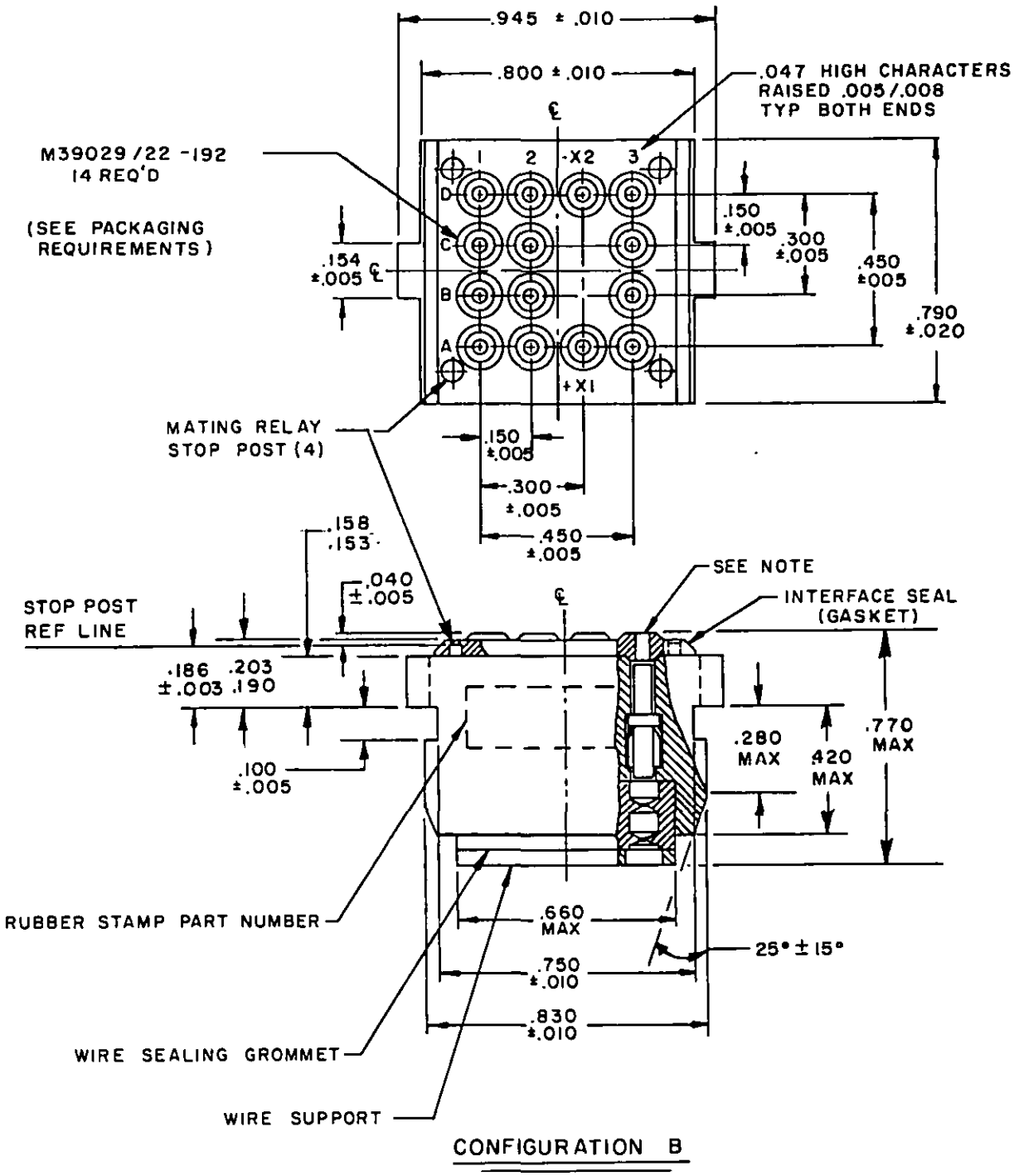


CONFIGURATION A

FIGURE 1. Socket configurations.

(D) denotes changes

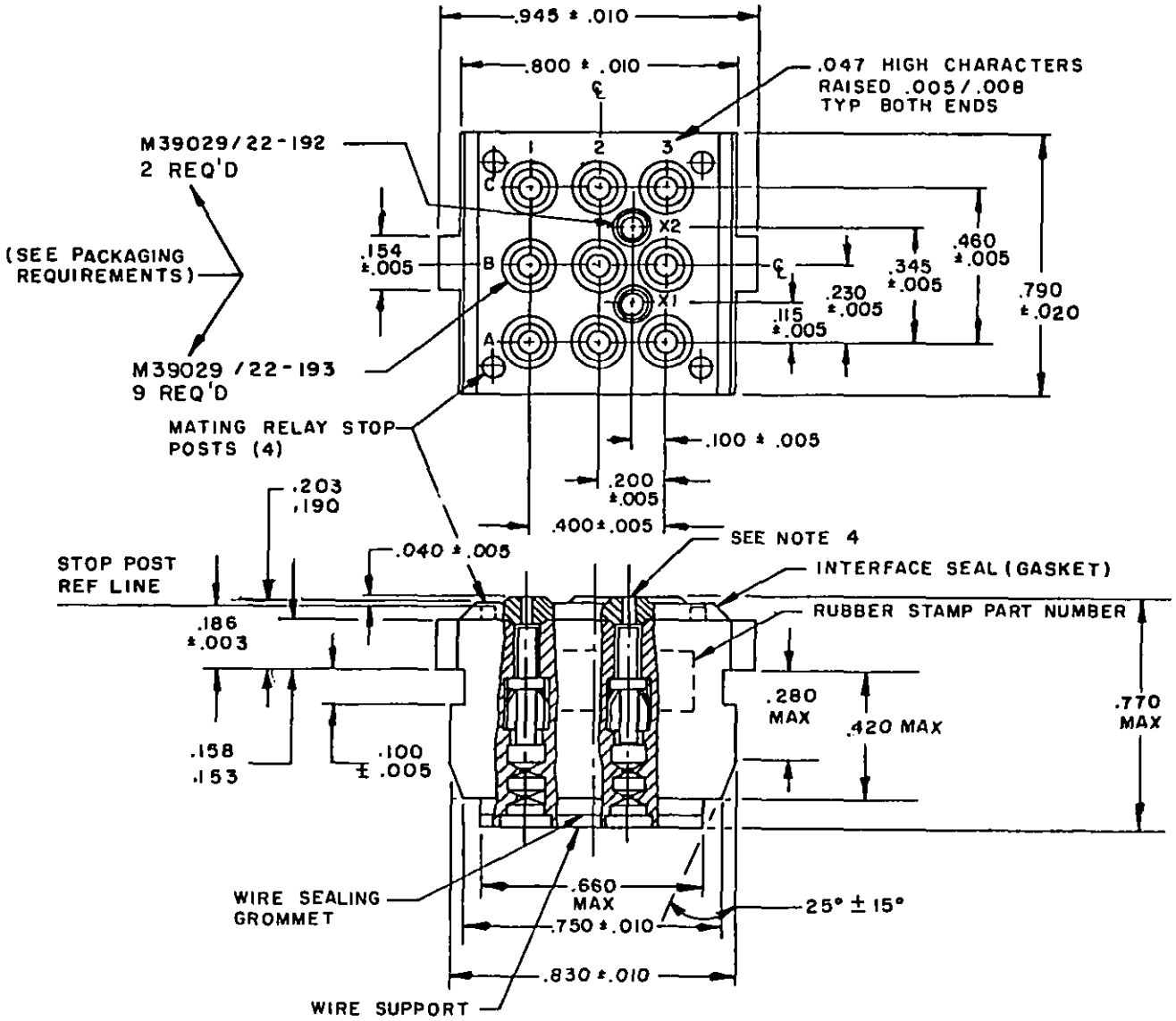
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CONFIGURATION B

FIGURE 1. Socket configurations - Continued.

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CONFIGURATION C

FIGURE 1. Socket configurations. - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.036	0.91	.190	4.83	.390	9.91
.005	0.13	.040	1.02	.203	5.16	.400	10.2
.008	0.20	.047	1.19	.265	6.73	.420	10.67
.010	0.25	.063	1.60	.280	7.11	.450	11.43
.020	0.51	.100	2.54	.290	7.37	.460	11.7
.025	0.64	.115	2.92	.300	7.62	.660	16.76
.030	0.76	.150	3.81	.345	8.76	.750	19.05
		.153	3.89	.375	9.52	.770	19.56
		.154	3.91			.790	20.1
		.158	4.01			.800	20.32
		.186	4.72			.830	21.08
						.945	24.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.03 (0.5 mm) for two place decimals.
4. Location of characters about their respective hole is optional both near and far side.
5. Material: Housing and wire support shall be a hard non-resilient, non-metallic material having electrical and mechanical characteristics capable of complying with the requirements of this specification. Grommet and interface seal shall be silicone rubber, color red.
6. Durometer of interface seal (gasket) is 40 ±5.
7. Wire sealing range:
   #20 .040 to .083.
   #16 .053 to .103.

FIGURE 1. Socket configurations. - Continued.

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REQUIREMENTS:

Applicable relay: See table I.

(D) Mating bracket: M6106/25-003.

Socket contacts: See table I and figure 1.

Tools: See table I and MIL-R-6106/31 and /32.

Insertion and withdrawal force: See table I.

(D) Altitude immersion: Applicable in accordance with MIL-S-12883.

Dielectric withstanding voltage (in socket module only, after installation of contacts, contact to contact):

At 25°C: 1500 V rms.
At 80,000 feet: 700 V rms.

Operating temperature range: -70°C to +125°C.

Weight including contacts:

Configuration A, 7 grams (.247 ounce).
Configuration B, 14 grams (.494 ounce).
Configuration C, 14 grams (.494 ounce).

Packaging: Socket assembly and contacts shall be packaged in separate packages within a common container.

Part number: M6106/23- (dash number from table I).

TABLE I. Dash number and characteristics.

Dash no.	Configuration	Applicable relay	Applicable socket contacts	Tools		Initial insertion force 1/ 2/	Initial withdrawal force 3/	
				No. 20 wire barrel	No. 16 wire barrel			
001	A	M6106/21-XXX, M83726/24-XXXX, or M83726/25-XXXX	M39029/22-192	M22520/7-01 crimping M22520/7-12 positioner	---	30 pounds max	9 pounds max	.35 pound min
002	B	M6106/22-XXX	M39029/22-192	MS27534-20 installing removal	---	45 pounds max	16 pounds max	.61 pound min
003	C	M6106/24-XXX	M39029/22-193 M39026/22-192		M22520/7-01 crimping M22520/7-13 positioner MS27534-16 installing removal	45 pounds max	19.3 pounds max	1.2 pounds min

1/ Insert mating relay in a straight, downward manner into the socket contained in the mounting bracket until relay is flush against socket face.
2/ To latch and lock relay against the socket, first apply a downward force on the relay end adjacent to one side of the bracket latches until the relay tab(s) are properly locked into the latch(es). Repeat the procedure to the opposite end of the relay to fully lock the relay in place.
3/ To release and remove the relay from the socket and the mounting bracket, use removal tool M6106/31-001.

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D TABLE II. Group A inspection.

Inspection	Requirements	Method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Examination of product 1/	As specified in MIL-R-6106	As specified in MIL-R-6106	1.0	2.5
Insertion force and withdrawal force 3/	Table I	As specified in MIL-S-12883	1.0	2/
Insulation resistance	As specified in MIL-R-6106	As specified in MIL-R-6106	1.0	2/
Dielectric withstanding voltage (sea level) 4/	As specified in MIL-R-6106	As specified in MIL-R-6106	1.0	2/

1/ Sample in accordance with MIL-STD-105 except, only two samples should be tested per lot for physical dimensions. If either of the two samples fails to meet the requirements, the complete lot shall then be inspected.

2/ Failures in this parameter are all considered major.

3/ In addition to the procedure outlined in MIL-S-12883, relays shall be removed from this socket using extraction tool M6106/31-001.

4/ Test shall be performed using a time duration of 2-5 seconds at a voltage level of 1500 V rms.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F633)

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